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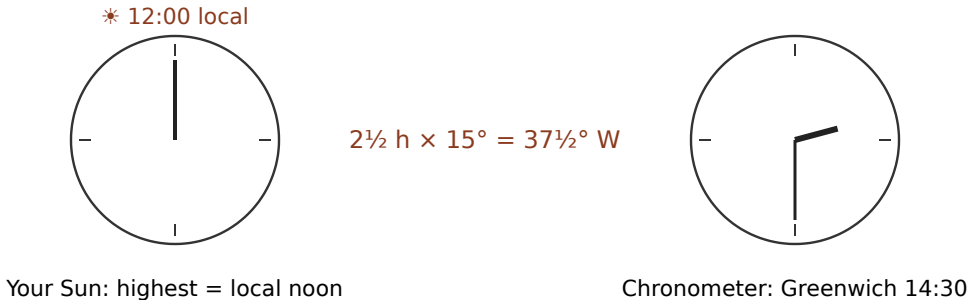
Longitude with a clock, in plain language

Before you sail, set an accurate chronometer to Greenwich time. Never reset it at sea. It is a piece of Greenwich you carry with you.

At sea, watch the Sun. Local noon is the moment it stands highest in your sky, not when an ordinary clock says 12. At that instant, read the chronometer.



Same instant, two times. Greenwich is ahead, so you are west.

Say it reads 14:30. Greenwich is 2 hours 30 minutes ahead of you. Earth turns 360° in 24 hours, so every hour is 15°. Two and a half hours is 37½° west. That number is your longitude. It is not how long you sailed. If Greenwich read 10:00 instead, you would be 30° east.

The 15° per hour holds at every latitude. What changes is the ground underneath: the circles you sail along (lines of latitude) shrink toward the poles, so the same 15° covers fewer miles. The clock still works everywhere.

What gets hard near the poles is the Sun, not the arithmetic. Its highest point is flatter and harder to pin down, and in polar winter it may not rise for weeks. At the pole itself every meridian meets, so longitude has no meaning.

What if we had started with tens?

We could divide a day into 10 hours, each hour into 100 minutes, and each minute into 100 seconds. A circle could have 400 divisions, called gradians, rather than 360 ordinary degrees. Revolutionary France actually tried decimal time. Tens fit our ten fingers and the way we write numbers.

Sixty has a useful advantage: it divides evenly by 2, 3, 4, 5, 6, 10 and 12. Ten does not. Decimal notation and modern calculation make that bookkeeping advantage less important, though one-third still cannot be written as a finite decimal.

A 400-day year would look tidy beside a 400-part circle, but nature does not oblige: the seasonal year is about 365.24 ordinary days. Changing clock units changes neither the day nor the year, and would not remove or reduce the need for leap-day corrections.

This simple example ignores the small seasonal difference between sundial time and clock time (the equation of time, up to about 16 minutes). Navigators correct for it with an almanac.

Royal Museums Greenwich, “What is longitude?” rmg.co.uk/stories/time/what-longitude · U.S. Naval Observatory, “The Equation of Time” aa.usno.navy.mil/faq/eqtime · Decimal time: IMCCE, “The decimal clock” · Calendar: USNO, “Leap Years”